

RDAS API Documentation

Base URL: `http://localhost:8080` (local) - all endpoints are under `/api/v1`. All responses are JSON. No authentication is required for the read APIs in this assessment build; the admin endpoints are flagged for auth in production.

Conventions

- **Pagination:** `page` (0-based, default 0), `size` (default 20, max 100).
- **Sorting:** `sort=field,dir` where `dir` is `asc` or `desc`. Repeatable. Sortable fields: `name`, `isoCode`, `capitalCity`, `continentCode`, `continentName`, `currencyCode`, `currencyName`. An unknown field returns 400.
- **Errors:** a consistent envelope (see end of document).

GET `/api/v1/countries`

Search and filter the country catalog. All filters are optional and combine with logical AND.

Query param	Description	Example
<code>name</code>	Partial, case-insensitive match on country name	<code>name=ken</code>
<code>continent</code>	Continent code or name	<code>continent=AF</code> or <code>continent=Africa</code>
<code>currency</code>	Currency ISO code or name	<code>currency=USD</code>
<code>language</code>	Language code or name	<code>language=French</code>
<code>page</code> , <code>size</code>	Pagination	<code>page=0&size=10</code>
<code>sort</code>	Sorting	<code>sort=name,desc</code>

Request

```
curl "http://localhost:8080/api/v1/countries?continent=AF&language=French&page=0&size=2&sort=name,asc"
```

Response `200 OK`

```
{
  "content": [
    {
      "isoCode": "BJ",
      "name": "Benin",
      "capitalCity": "Porto Novo",
      "continentCode": "AF",
      "continentName": "Africa",
      "currencyCode": "XOF",
      "currencyName": "Francs",
      "flagUrl": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Benin.jpg"
    },
    {
      "isoCode": "BF",
      "name": "Burkina Faso",
      "capitalCity": "Ouagadougou",
      "continentCode": "AF",
      "continentName": "Africa",
      "currencyCode": "XOF",
      "currencyName": "Francs",
      "flagUrl": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Burkina_Faso.jpg"
    }
  ],
  "page": 0,
  "size": 2,
  "totalElements": 16,
  "totalPages": 8,
  "first": true,
  "last": false,
  "sort": "name: ASC"
}
```

GET /api/v1/countries/{isoCode}

Full detail for a single country, including the languages spoken.

`isoCode` is a 2-3 letter country code (case-insensitive).

Request

```
curl http://localhost:8080/api/v1/countries/KE
```

Response 200 OK

```
{
  "isoCode": "KE",
  "name": "Kenya",
  "capitalCity": "Nairobi",
  "phoneCode": "254",
  "continentCode": "AF",
  "continentName": "Africa",
  "currencyCode": "KES",
  "currencyName": "Shillings",
  "flagUrl": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Kenya.jpg",
  "languages": [ { "code": "swa", "name": "Swahili" } ],
  "sourceRefreshedAt": "2026-06-11T11:40:12.308374Z"
}
```

Errors: 404 if no such country; 400 if `isoCode` is not a 2-3 letter code.

GET /api/v1/countries/{isoCode}/currency-peers

Countries that use the same currency as the given country (the given country included), sorted by name.

Request

```
curl http://localhost:8080/api/v1/countries/US/currency-peers
```

Response 200 OK

```
[
  { "isoCode": "EC", "name": "Ecuador", "capitalCity": "Quito",
    "continentCode": "AM", "continentName": "The Americas",
    "currencyCode": "USD", "currencyName": "Dollars", "flagUrl": "http://.../Ecuador.jpg"
  },
  { "isoCode": "US", "name": "United States", "capitalCity": "Washington",
    "continentCode": "AM", "continentName": "The Americas",
    "currencyCode": "USD", "currencyName": "Dollars", "flagUrl":
      "http://.../United_States.jpg" }
]
```

Reference lists

GET /api/v1/continents

```
curl http://localhost:8080/api/v1/continents
```

```
[ { "code": "AF", "name": "Africa" }, { "code": "AS", "name": "Asia" } ]
```

GET /api/v1/currencies

```
[ { "code": "KES", "name": "Shillings" }, { "code": "USD", "name": "Dollars" } ]
```

GET /api/v1/languages

```
[ { "code": "swa", "name": "Swahili" }, { "code": "eng", "name": "English" } ]
```

GET /api/v1/admin/catalog/status

```
{
  "countryCount": 244,
  "continentCount": 6,
  "currencyCount": 174,
  "languageCount": 407,
  "lastRefreshStartedAt": "2026-06-11T11:39:57.263Z",
  "lastRefreshCompletedAt": "2026-06-11T11:42:26.696Z",
  "lastRefreshSuccessful": true,
  "lastError": null,
  "refreshInProgress": false,
  "stale": false
}
```

POST /api/v1/admin/catalog/refresh

Triggers an asynchronous full refresh. Returns `202 Accepted` with the current status. Concurrent triggers are coalesced (single-flight).

```
curl -X POST http://localhost:8080/api/v1/admin/catalog/refresh
```

Health and metrics

Endpoint	Purpose
GET /actuator/health	Aggregate health (catalog, db, circuit breaker).
GET /actuator/health/readiness	Kubernetes readiness (db + readinessState; SOAP excluded).
GET /actuator/health/liveness	Kubernetes liveness.
GET /actuator/metrics	Micrometer metrics.
GET /actuator/prometheus	Prometheus scrape (HTTP + Resilience4j metrics).

Error envelope

Every handled error returns this structure with the appropriate HTTP status:

```
{
  "timestamp": "2026-06-11T12:00:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid sort property 'hack'. Allowed: [name, isoCode, ...]",
  "path": "/api/v1/countries",
  "fieldErrors": [ { "field": "name", "message": "name filter must be at most 100
                    characters" } ]
}
```

Status	When
400	Validation failure, bad sort field, bad parameter type, malformed ISO code
404	Country not found
502	SOAP integration failure (only reachable via the admin/harvest path)
500	Unexpected server error (no stack trace leaked)